Benjamin Abensur

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EDUCATION

EPFL — Swiss Federal Institute of Technology in Lausanne Expected Graduation: Jan 2028

Bachelor of Science in Communication Systems, School of Computer and Communication Sciences (IC)

- Relevant Coursework: Introduction to Machine Learning (**5.75/6**), Algorithms I (**5.5/6**), Linear Algebra (**5.5/6**), Probability & Statistics, Numerical Methods for Visual Computing & ML, Computer Architecture, Computer Systems, Responsible Software

WORK EXPERIENCE

Biosynex, *AI Engineer (part-time);* Remote *(Euronext-listed rapid diagnostics)* Oct 2026 – Present

- Building a RAG-based copilot over regulatory and quality documentation to speed up compliance and document search.
- Deploying AI agents and automation pipelines (tender responses, reporting, recurring tasks) and leading AI adoption through training on LLMs, agentic workflows and automation.

EPFL, *Teaching Assistant, Mechanics (PHYS-101) and Linear Algebra (MATH-111)* Sep 2026 – Present

- Selected on academic merit for a Mechanics course of **1,000+** students (CS, Chemistry, EE); lead a weekly 2-hour exercise session and answer questions on the course's Ed forum.

NanoSynex, *Software Engineer, ML & Statistical Modeling;* Remote *(Technion medtech spin-off)* Jun – Sep 2024

- Benchmarked the rapid antibiotic test against a veterinary reference laboratory for a validation study, merging both result sets by sample ID in pandas and quantifying agreement rates.
- Modeled bacterial growth in Python and NumPy to compare Wald's sequential probability ratio test (SPRT) with a fixed-threshold rule for antibiotic susceptibility testing. In simulation, the SPRT reached a decision **20%** sooner, with a major-error rate below **3%**.
- Deployed two offline LLMs behind a model router with shared memory, enabling analysis of confidential R&D and investor documents without any data leaving the machine.
- Automated inbox triage in Python with an LLM classifier routing email into custom priority categories (investors, grants, R&D).

PROJECTS

ReCHor — public-transport journey planner, CS-108 github.com/Benabens/ReCHor

Java, JavaFX

- Finds every Pareto-optimal journey between two Swiss stops (Connection Scan Algorithm) over real CFF timetables (~258 MB).
- Memory-mapped, bit-packed timetable storage; JavaFX UI with autocomplete and iCalendar export; 44 classes, ~3,600 lines of tests.

Kairo — internship search and application platform, live with public users since Oct 2026 kairocareer.app

Next.js, TypeScript, Python, PostgreSQL (Supabase, row-level security), GitHub Actions, Railway, Vercel

- Scheduled Python pipeline on Railway pulls openings from employers' public job feeds and ATS APIs, deduplicates them, then an LLM worker scores each one against every user's profile.
- Multi-tenant Postgres isolated by row-level security; application copilot drafts each form answer with its source and flags uncertain ones for the user to check before submitting; 160+ tests, CI.
- Referral finder: LLM extraction turns freeform notes into structured contacts, then links every opening to the user's direct and indirect connections at that company, so no warm intro is missed.

Cortex — exam-preparation platform cortexexam.com · GitHub

Next.js, TypeScript, PostgreSQL, Drizzle, LaTeX, Railway

- Learns each course's exam format from past papers, weights every topic by how heavily it was examined, schedules spaced revision on weak points, and generates faithful practice exams via a multi-pass LLM pipeline; used it to prepare my own exams in **3** EPFL courses.
- Answers accepted only if verified by sandboxed code execution or symbolic checks; per-user Postgres schemas, cost caps; 330+ tests.

Gaming Addiction Prediction — ML from scratch, CS-233 *(team of 3)* github.com/Benabens/addiction-classifier-numpy

Python, NumPy, no ML libraries

- Predicted gaming-addiction level (3 classes) and score (0–10) from gaming and mental-health data with KNN, logistic/linear regression, K-Means and an MLP with hand-written backpropagation; two written reports.
- Inverse-frequency weighted cross-entropy lifted rarest-class recall from **0 to 0.69** and macro-F1 from **0.55 to 0.76** (5-fold CV).

ICoop — two-player cooperative 2D game, CS-107 github.com/Benabens/ICoop

Java

- Fire-and-water co-op game with a boss fight; interactions via double dispatch (visitor pattern). ~3,700 lines.

Other Projects: Verilog digital-logic library (42/42 self-checking testbenches); networking and UNIX-filesystem project in C (CS-202, in a pair); *VeriReason*, a PyTorch Transformer from first principles (in progress); Reed-Solomon QR-code generator (Java, CS-107).

LEADERSHIP & COMMUNITY INVOLVEMENT

Scout Summer Camp, *Co-Camp Director;* Mévouillon, France Jul 2025

- Co-directed a one-month camp for **80** children from two local groups (Nice, Strasbourg); led a team of **15** counsellors, promoted team leads and wrote the camp's full educational project.
- Co-managed the **€80k** camp budget, running the €18k food line myself (local farmers, Intermarché); built the site at pre-camp: tents, dry toilets, wooden tables; organised outings and explorations.

EPFL Linear Algebra Bootcamp (MATH-111), *Co-instructor;* Lausanne, Switzerland Nov 2025

- Gave the lectures and wrote the exercise sets of a paid one-week bootcamp for **30** first-year students (mid-semester break).

BABOO, *Founder & Event Producer;* Lausanne, Switzerland 2025

- Produced **3** club nights end to end that brought in **CHF 10,000**, one at Noche Club with an internationally touring DJ.

SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, Verilog, x86/RISC-V assembly
ML & AI: PyTorch, NumPy, pandas, matplotlib, Ollama
Web & Systems: Next.js, React, Node.js, REST APIs, PostgreSQL, Supabase, web scraping, GitHub Actions, Linux, POSIX sockets, Git
Spoken: French (native), English (fluent), Hebrew (basic), Spanish (basic)